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CANDIDATE NUMBER

SYDNEY GRAMMAR SCHOOL



2015 Assessment Examination

FORM VI

MATHEMATICS EXTENSION I

Monday 18th May 2015

General Instructions

- Writing time — 2 hours
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Total — 70 Marks

- All questions may be attempted.

Section I – 10 Marks

- Questions 1–10 are of equal value.
- Record your solutions to the multiple choice on the sheet provided.

Section II – 60 Marks

- Questions 11–14 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each answer booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Write your candidate number on this question paper and hand it in with your answers.
- Place everything inside the answer booklet for Question Eleven.

Checklist

- SGS booklets — 4 per boy
- Multiple choice answer sheet
- Candidature — 112 boys

Examiner

LYL

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

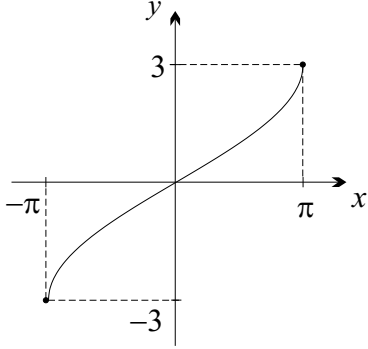
QUESTION ONE

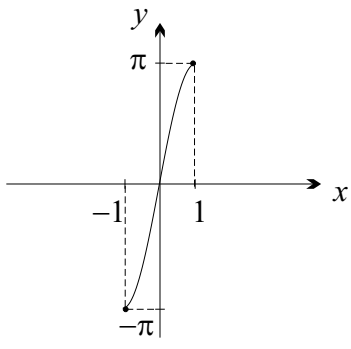
The polynomial $P(x) = x^4 + x^3 - 7x^2 - x + 6$ has four linear factors. Which expression below is NOT a factor of $P(x)$?

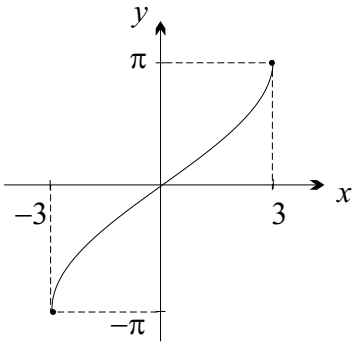
- (A) $x - 1$
- (B) $x - 2$
- (C) $x + 1$
- (D) $x + 6$

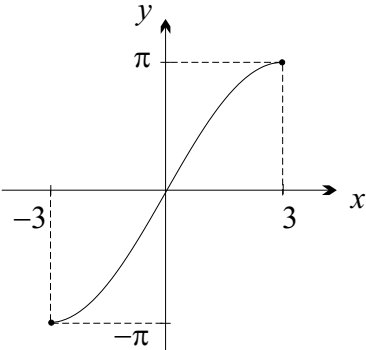
QUESTION TWO

Which graph best represents $y = 2 \sin^{-1} \frac{x}{3}$?

- (A) 

(B) 

(C) 

(D) 

QUESTION THREE

A particle is moving in simple harmonic motion according to the equation

$$x = 1 + 6 \sin\left(2t + \frac{5\pi}{6}\right).$$

In what interval does the particle oscillate?

- (A) $-6 \leq x \leq 6$
- (B) $-5 \leq x \leq 7$
- (C) $5 \leq x \leq 7$
- (D) $-1 \leq x \leq 3$

QUESTION FOUR

The exact value of $\sec\left(\sin^{-1}\left(-\frac{1}{3}\right)\right)$ is:

- (A) $-\frac{3\sqrt{2}}{4}$
- (B) $\frac{3\sqrt{2}}{4}$
- (C) $\frac{2\sqrt{2}}{3}$
- (D) $-\frac{2\sqrt{2}}{3}$

QUESTION FIVE

The expression $\sqrt{3}\sin x - \cos x$ is equivalent to:

- (A) $2\sin\left(x - \frac{\pi}{6}\right)$
- (B) $2\sin\left(x + \frac{\pi}{6}\right)$
- (C) $2\sin\left(x + \frac{5\pi}{6}\right)$
- (D) $2\sin\left(x - \frac{5\pi}{6}\right)$

QUESTION SIX

Which of the following is an expression for $\int \cos^2 2x \, dx$?

- (A) $\frac{x}{2} - \frac{1}{8} \sin 4x + C$
- (B) $x - \frac{1}{4} \sin 4x + C$
- (C) $\frac{x}{2} + \frac{1}{8} \sin 4x + C$
- (D) $x + \frac{1}{4} \sin 4x + C$

QUESTION SEVEN

A particle moves on a horizontal line so that its displacement x cm from the origin is given by $x = t^3 - 5t^2 - 3t + 4$. Take right as the positive direction.

At time $t = 2$ seconds the particle is:

- (A) right of the origin, travelling to the left and accelerating to the right
- (B) left of the origin, travelling to the left and accelerating to the right
- (C) left of the origin, travelling to the right and accelerating to the right
- (D) right of the origin, travelling to the right and accelerating to the left

QUESTION EIGHT

Consider the function $f(x) = \sin x + \frac{1}{2} \cos 2x$ in the interval $0 \leq x \leq 2\pi$. Which of the following is the x -coordinate of a stationary point of $f(x)$?

- (A) 0
- (B) $\frac{\pi}{6}$
- (C) $\frac{7\pi}{6}$
- (D) $\frac{11\pi}{6}$

QUESTION NINE

A projectile is fired with an initial velocity of 30 m/s at an angle of elevation of 60° to the horizontal. Let x and y be the respective horizontal and vertical components of the displacement from the point of projection, and take $g = 10 \text{ m/s}^2$.

The initial conditions are:

- (A) $\dot{x} = 15\sqrt{3}$, $\ddot{x} = -10$, $\dot{y} = 15$ and $\ddot{y} = 0$
- (B) $\dot{x} = 30\sqrt{3}$, $\ddot{x} = 0$, $\dot{y} = 30\sqrt{3}$ and $\ddot{y} = -10$
- (C) $\dot{x} = 15\sqrt{3}$, $\ddot{x} = 0$, $\dot{y} = 15$ and $\ddot{y} = -10$
- (D) $\dot{x} = 15$, $\ddot{x} = 0$, $\dot{y} = 15\sqrt{3}$ and $\ddot{y} = -10$

QUESTION TEN

Find the values of x for which $(x + 1)(x - 2)(x - 3) > 0$.

- (A) $-1 < x < 2$ or $x > 3$
- (B) $x < -1$ or $2 < x < 3$
- (C) $x > -1$ or $2 < x < 3$
- (D) $-1 < x < 3$

————— End of Section I —————

Exam continues overleaf ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

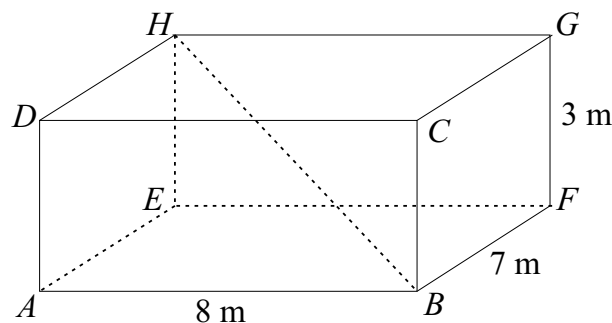
Start a new booklet for each question.

QUESTION ELEVEN (15 marks) Use a separate writing booklet.

Marks

- (a) Find the exact value of $\int_0^{\frac{1}{6}} \frac{1}{\sqrt{1-9x^2}} dx$. 2
- (b) Find the value of k if $x - 2$ is a factor of $P(x) = x^3 - 3kx + 10$. 2
- (c) The equation $x^3 + 3x^2 - 2x + 1 = 0$ has roots α , β and γ .
- (i) Write down the values of $\alpha\beta + \alpha\gamma + \beta\gamma$ and $\alpha\beta\gamma$. 2
- (ii) Hence, evaluate $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$. 1

(d)



The diagram above shows a rectangular prism with dimensions 8 m by 7 m by 3 m. Find the angle the diagonal BH makes with the base $ABFE$. Leave your answer correct to the nearest minute. 2

- (e) Without the aid of calculus, sketch the graph of $P(x) = (x + 1)(x - 2)(x + 3)^3$. Show clearly any features including all intercepts with the axes. 2
- (f) Initially a ball is thrown at 20 m/s at an angle of elevation of 30° from the top of a building 40 m high. The equations of motion of the ball are

$$x = 10t\sqrt{3}$$

$$y = -5t^2 + 10t$$

where x and y are the horizontal and vertical components of displacement from the point of projection at time t seconds after the ball is thrown.

- (i) At what time will the ball hit the ground? 2
- (ii) Find the horizontal range of its flight. 1
- (iii) Find the Cartesian equation of its path. 1

Exam continues next page ...

QUESTION TWELVE (15 marks) Use a separate writing booklet.

Marks

- (a) (i) Factorise $x^3 - 5x^2 + 8x - 4$.

2

- (ii) Hence solve the equation $x^3 - 5x^2 + 8x - 4 = 0$.

1

- (b) The function $f(x) = 3 - \sqrt{x - 2}$ is defined over the domain $x \geq 2$.

2

Find the equation of the inverse function $f^{-1}(x)$ and state its domain.
- (c) Prove that $\sin(\theta + \frac{\pi}{3})\sin(\theta - \frac{\pi}{3}) = \sin^2 \theta - \frac{3}{4}$.

2

- (d) The acceleration of a particle P is given by $\ddot{x} = -2e^{-x}$ where x is the displacement from the origin O and right is taken as the positive direction. The particle starts at the origin with a velocity of 2 m/s.
- (i) Show that $v^2 = 4e^{-x}$.

1

- (ii) Assuming that v is positive, find the displacement as a function of time.

2

- (iii) Briefly describe the displacement and velocity of the particle as $t \rightarrow \infty$.

1

- (iv) Explain why the velocity could be assumed to be positive in part (ii).

1

- (e) (i) Differentiate $y = x \tan^{-1} x$.

1

- (ii) Hence find a primitive of $\tan^{-1} x$.

1

- (iii) Find the area bounded by the curve $y = \tan^{-1} x$, the x -axis and the line $x = 1$.

1

QUESTION THIRTEEN (15 marks) Use a separate writing booklet.**Marks**

- (a) Two zeroes of a polynomial $P(x)$ of degree 3 are -2 and -4 . When $x = 1$ it takes the value of 15 and when $x = -3$ it takes the value of 7. Find the polynomial. **3**

- (b) The motion of a particle is given by $x = 1 + 2 \sin 3t$, where $t \geq 0$.

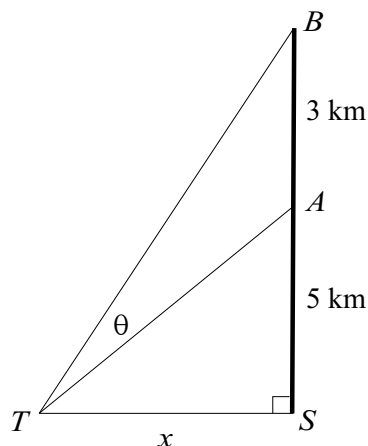
- (i) Prove that the motion is simple harmonic by showing that **1**

$$\ddot{x} = -n^2(x - x_0).$$

- (ii) Write down the period and the amplitude of the motion. **2**

- (iii) Find the first two times when the particle returns to the centre of motion. Give your answers as exact values. **2**

(c)

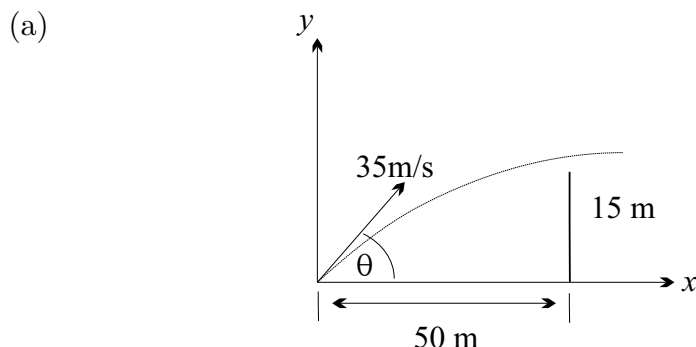


The diagram above shows a technician T observing work on a pipeline that is being built out from the shores of Darwin. The technician is standing onshore x km due west of the start of the pipeline S . He can see two company boats A and B which are respectively 5 km and 8 km due north of the point S . Let θ be the angle that AB subtends at T .

- (i) Show that $\theta = \tan^{-1} \frac{8}{x} - \tan^{-1} \frac{5}{x}$. **1**

- (ii) Show that θ is maximised when the technician is $2\sqrt{10}$ km from the point S . **4**

- (iii) Hence show that the maximum value of θ is $\tan^{-1} \left(\frac{3}{4\sqrt{10}} \right)$. **2**

QUESTION FOURTEEN (15 marks) Use a separate writing booklet.**Marks**

The diagram above shows a stone thrown with a velocity of 35 m/s at an angle θ so that it just clears a 15 m high wall that is 50 m from the origin on the same horizontal plane.

- (i) Show that the two equations of motion for the horizontal and vertical components of displacement are respectively:

2

$$x = 35t \cos \theta$$

$$y = 35t \sin \theta - 5t^2$$

Assume that $g = 10 \text{ m/s}^2$ and t is time in seconds.

- (ii) Find the angles at which the stone could be thrown.
Give your answers correct to the nearest degree.

3

- (b) A line with gradient m intersects the cubic curve $y = (x - 1)(x + 2)(x - 3)$ at the point $P(3, 0)$ and at two other points Q and R .

- (i) Show that the x coordinates of the points of intersection satisfy the equation:

1

$$x^3 - 2x^2 - (m + 5)x + 6 + 3m = 0$$

- (ii) Find the equation of the line through P which is also a tangent to the curve at another distinct point.

3

- (c) On a certain day the depth of water in a harbour is 3 metres at low tide and 9 metres at high tide. Low tide occurs at 5:00 am and the following high tide at 1:00 pm. Assume the rise and fall of the tides is simple harmonic. Find between what times on that day a ship may safely enter the harbour, if a minimum depth of 4 metres of water is required.

3

- (d) (i) Find the general solutions of the equation $2 \sin 3x \cos 4x - 1 = \cos 4x - 2 \sin 3x$.
(ii) Hence find the two smallest positive solutions of this equation.

2**1**

————— End of Section II —————

END OF EXAMINATION

B L A N K P A G E

B L A N K P A G E

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

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MATHEMATICS EXTENSION I
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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Question Nine

A ☐ B ☐ C ☐ D ☐

Question Ten

A ☐ B ☐ C ☐ D ☐

Form II

Ext 1 Assessment 2015

Q11 a) $\int_0^{\frac{1}{3}} \frac{1}{\sqrt{1-9x^2}} dx = \frac{1}{3} \int_0^{\frac{1}{3}} \frac{1}{\sqrt{\frac{1}{9}-x^2}} dx$

$$= \frac{1}{3} \left[\sin^{-1} 3x \right]_0^{\frac{1}{3}} \checkmark$$

$$= \frac{1}{3} \left(\sin^{-1} \frac{1}{2} - 0 \right)$$

$$= \frac{\pi}{18} \checkmark$$

b) $P(2) = 8 - 6k + 10 \checkmark$ $P(2) = 0$

$$18 - 6k = 0$$

$$-6k = -18$$

$$k = 3 \checkmark$$

c) i) $xp + xr + pr = -2 \checkmark$

$$xpr = -1 \checkmark$$

ii) $\frac{1}{x} + \frac{1}{p} + \frac{1}{r} = \frac{xp + xr + pr}{xpr}$

$$= 2 \checkmark$$

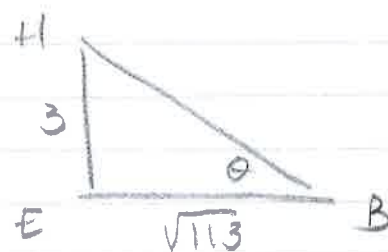
d) $DE^2 = AE^2 + AD^2$

$$= 8^2 + 7^2$$

$$= 113$$

$DE = \sqrt{113} \checkmark$

$\triangle HEB$ right angled



$\tan \theta = \frac{3}{\sqrt{113}}$

$\theta \doteq 15^\circ 46'$ (nearest minute) $\checkmark$

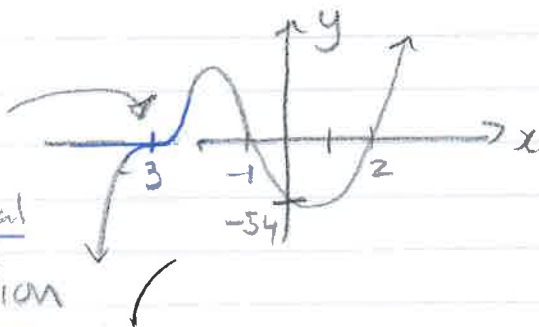
e)

$x = -3$
triple root

horizontal
point of
inflexion

- clear

concavity change



$x = -1$
 $x = 2$ } single roots.

intercepts $\checkmark$

shape and single/triple roots must be clearly shown

MC

Q1 D

Q2 C

Q3 B

Q4 B

Q5 A

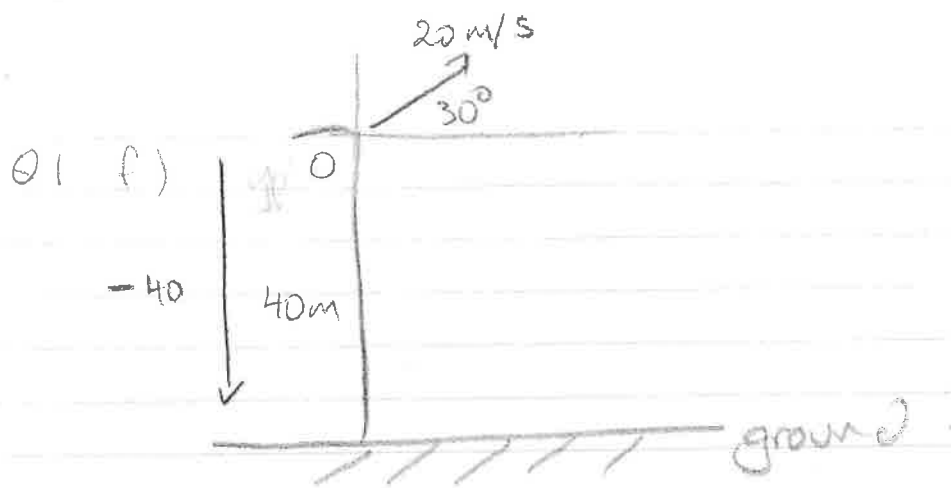
Q6 C

Q7 B

Q8 B

Q9 D

Q10 A



i) $t = 0$ $y = 0$

Ground is at $y = -40$

$$-5t^2 + 10t = -40$$

$$5t^2 - 10t + 40 = 0$$

$$t^2 - 2t - 8 = 0$$

$$(t + 2)(t - 4) = 0$$

$$t = -2 \text{ or } 4$$

$$t = 4 \text{ seconds } (t > 0)$$

ii) $x = 10 \times 4 \times \sqrt{3}$
 $= 40\sqrt{3}$

iii) $t = \frac{x}{10\sqrt{3}}$ $y = -5 \left(\frac{x}{10\sqrt{3}} \right)^2 + 10 \left(\frac{x}{10\sqrt{3}} \right)$

$$= -\frac{5x^2}{100 \times 3} + \frac{x}{10\sqrt{3}}$$

$$y = -\frac{x^2}{60} + \frac{x\sqrt{3}}{3}$$

$$= \frac{20x\sqrt{3} - x^2}{60}$$

Q12 a) i) let $P(x) = x^3 - 5x^2 + 8x - 4 = 0$

Test $P(1) = 1 - 5 + 8 - 4$
 $= -4 + 8 - 4$
 $= 0$

$P(2) = 8 - 5 \times 4 + 8 \times 2 - 4$
 $= 8 - 20 + 16 - 4$
 $= 0$

$(x-1)$ and $(x-2)$ are factors

✓ at least one factor

$1 + 2 + \alpha = 5$ (α is the third root)
 $\alpha = 2$

$P(x) = (x-1)(x-2)^2$ ✓

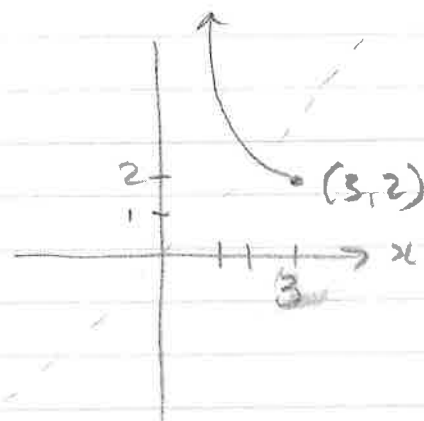
long div or other methods ok

ii) $(x-1)(x-2)^2 = 0$
 $x = 1$ or $x = 2$ ✓

b) $f(x) = 3 - \sqrt{x-2}$
 Swap x and y and rearrange
 $x = 3 - \sqrt{y-2}$
 $x - 3 = -\sqrt{y-2}$

$(x-3)^2 = y-2$
 $y = (x-3)^2 + 2$ ✓

Domain $x \leq 3$ ✓



c) LHS = $\sin\left(\theta + \frac{\pi}{3}\right) \sin\left(\theta - \frac{\pi}{3}\right)$
 $= \left(\sin\theta \cos\frac{\pi}{3} + \sin\frac{\pi}{3} \cos\theta\right) \left(\sin\theta \cos\frac{\pi}{3} - \sin\frac{\pi}{3} \cos\theta\right)$
 $= \left(\frac{\sin\theta}{2} - \frac{\sqrt{3}\cos\theta}{2}\right) \left(\frac{\sin\theta}{2} - \frac{\sqrt{3}\cos\theta}{2}\right)$ ✓
 $= \frac{\sin^2\theta}{4} - \frac{3\cos^2\theta}{4}$
 $= \frac{\sin^2\theta - 3(1 - \sin^2\theta)}{4}$ ✓
 $= \frac{4\sin^2\theta - 3}{4}$
 $= \sin^2\theta - \frac{3}{4}$ as required.

Q12 d) i) $\ddot{x} = -2e^{-x}$

$$\frac{d}{dx}\left(\frac{1}{2}v^2\right) = -2e^{-x}$$

$$\frac{1}{2}v^2 = 2e^{-x} + \frac{1}{2}C$$

$$v^2 = 4e^{-x} + C$$

when $x=0$ $v=2$

$$4 = 4e^{-0} + C$$

$$C=0$$

$$v^2 = 4e^{-x}$$

ii) At $x=0$ $v=2$ given $v>0$

$$v = 2e^{-\frac{x}{2}}$$

$$\frac{dx}{dt} = \frac{2}{e^{\frac{x}{2}}}$$

$$\frac{dt}{dx} = \frac{e^{\frac{x}{2}}}{2}$$

integrate

$$t = \frac{1}{2} \times 2e^{\frac{x}{2}} + D$$

$$t = e^{\frac{x}{2}} + D$$

$t=0$ $x=0$

$$0 = 1 + D$$

$$D = -1$$

$$t = e^{\frac{x}{2}} - 1$$

$$e^{\frac{x}{2}} = t + 1$$

$$\frac{x}{2} = \ln(t+1)$$

$$x = 2\ln(t+1)$$

iii) The particle starts at the origin and moves right. As $t \rightarrow \infty$, $x \rightarrow \infty$, $v \rightarrow 0^+$, ie slowing down as $t \rightarrow \infty$.

iv) Initially v is positive.

$t=0$ $x=0$ $v=2$

$t>0$ $x>0$ so v can never equal zero.

Hence v can never be negative.

012e) i) $y = x \tan^{-1} x$
 $= uv$

$u = x$
 $u' = 1$

$v = \tan^{-1} x$
 $v' = \frac{1}{1+x^2}$

$y' = \frac{x}{1+x^2} + \tan^{-1} x \checkmark$

ii) $\int \tan^{-1} x \, dx + \int \frac{x}{1+x^2} \, dx = x \tan^{-1} x$

$\int \tan^{-1} x \, dx = x \tan^{-1} x - \frac{1}{2} \int \frac{2x}{1+x^2} \, dx$

$= x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) + C \checkmark$

iii) $\int_0^1 \tan^{-1} x \, dx$

$= \left[x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) \right]_0^1$

$= (\tan^{-1} 1 - \frac{1}{2} \ln 2) - 0$

$= \left(\frac{\pi}{4} - \frac{1}{2} \ln 2 \right) \checkmark$

$= \frac{\pi - 2 \ln 2}{4} \text{ units}^2$

Q13 a) $P(x) = (x+2)(x+4)(ax+b)$
since $P(x)$ degree 3 a & b constants.

$$P(1) = 15$$

$$3 \times 5 \times (a+b) = 15$$

$$a+b = 1$$

(1)

✓

$$P(-3) = 7$$

$$-1 \times 1 \times (-3a+b) = 7$$

$$-3a+b = -7$$

(2)

(1) - (2)

$$4a = 8$$

$$a = 2$$

$$b = -1$$

}

✓

$$P(x) = (x+2)(x+4)(2x-1)$$

✓

b) i)

$$x = 1 + 2 \sin 3t$$

$$\dot{x} = 6 \cos 3t$$

$$\ddot{x} = -18 \sin 3t$$

$$= -9 (1 + 2 \sin 3t - 1)$$

$$= -3^2 (x - 1)$$

which is in the form

$$= -n^2 (x - x_0)$$

which proves the motion is SHM.

✓

ii)

period

$$n = 3$$

$$T = \frac{2\pi}{3}$$

✓

amplitude = 2

✓

iii)

centre of motion

$$x_0 = 1$$

$$1 = 1 + 2 \sin 3t$$

$$2 \sin 3t = 0$$

$$\sin 3t = 0$$

$$3t = \pi \text{ or } 2\pi$$

$$t = \frac{\pi}{3} \text{ or } \frac{2\pi}{3}$$

✓

✓

Q13 c) i) $\triangle ATS$ right angled

let $\angle ATS = \alpha$

$$\tan \alpha = \frac{5}{x}$$

$$\alpha = \tan^{-1} \frac{5}{x}$$

$\triangle BTS$ right angled

$$\tan(\alpha + \theta) = \frac{8}{x}$$

$$\alpha + \theta = \tan^{-1} \frac{8}{x}$$

$$\alpha + \theta - \alpha = \theta$$

$$\theta = \tan^{-1} \frac{8}{x} - \tan^{-1} \frac{5}{x}$$

must show some working to derive θ .

$$ii) \theta = \tan^{-1} \frac{8}{x} - \tan^{-1} \frac{5}{x}$$

$$\frac{d\theta}{dx} = \frac{-8}{x^2 + 64} + \frac{5}{x^2 + 25}$$

when $\frac{d\theta}{dx} = 0$

$$\frac{5}{x^2 + 25} - \frac{8}{x^2 + 64} = 0$$

$$\frac{5(x^2 + 64) - 8(x^2 + 25)}{(x^2 + 25)(x^2 + 64)} = 0$$

$$5x^2 + 5 \times 64 - 8x^2 - 200 = 0$$

$$-3x^2 + 120 = 0$$

$$3x^2 = 120$$

$$x^2 = 40$$

$$x = \sqrt{40}$$

$$= 2\sqrt{10}$$

$$\frac{d}{dx} \tan^{-1}(8x^{-1})$$

let $u = 8x^{-1}$

$$\frac{du}{dx} = -8x^{-2}$$

$$y = \tan^{-1} u$$

$$\frac{dy}{du} = \frac{1}{1+u^2}$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$= \frac{1}{1 + \left(\frac{8}{x}\right)^2} \times -\frac{8}{x^2}$$

$$= \frac{-8}{x^2 + 64}$$

Must show maximum with table of values.

x	5	$2\sqrt{10}$	7
$\frac{d\theta}{dx}$	≈ 0.010	0	≈ 0.0032
	$\uparrow$ 45		
	89 x 50		

$$\frac{5(7^2 + 64) - 8(7^2 + 25)}{113 \times 75}$$

must provide working/values for marks.

$$Q13 \text{ iii) } \theta = \tan^{-1} \frac{8}{2\sqrt{10}} - \tan^{-1} \frac{5}{2\sqrt{10}}$$

$$\text{let } \alpha = \tan^{-1} \frac{8}{2\sqrt{10}} \quad \beta = \tan^{-1} \frac{5}{2\sqrt{10}}$$

$$\tan \alpha = \frac{8}{2\sqrt{10}}$$

$$\tan \beta = \frac{5}{2\sqrt{10}}$$

$$\theta = \alpha - \beta$$

$$\tan \theta = \tan(\alpha - \beta)$$

$$= \frac{\tan \alpha - \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$= \frac{\frac{8}{2\sqrt{10}} - \frac{5}{2\sqrt{10}}}{1 + \frac{8 \times 5}{2\sqrt{10} \times 2\sqrt{10}}}$$

$$= \frac{\frac{3}{2\sqrt{10}} \times \frac{1}{2}}{\frac{3}{4\sqrt{10}}}$$

$$= \frac{3}{4\sqrt{10}}$$

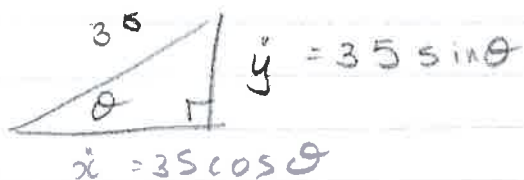
$$\text{so } \tan \theta = \frac{3}{4\sqrt{10}}$$

$$\theta = \tan^{-1} \frac{3}{4\sqrt{10}} \text{ as required}$$

FORM VI

ASSESSMENT 2015

Q14 a) i)



$$\ddot{x} = 0$$

integrate wrt x

$$\dot{x} = C_1$$

when $t=0$ $\dot{x} = 35 \cos \theta$ $C_1 = 0$

$$\dot{x} = 35 \cos \theta$$

integrate

$$x = 35t \cos \theta + C_2$$

when $t=0$ $x=0$ $C_2=0$

$$x = 35t \cos \theta$$

✓

most
derive
equations.

$$\ddot{y} = -10$$

integrate wrt x

$$\dot{y} = -10t + C_3$$

when $t=0$ $\dot{y} = 35 \sin \theta$

$$C_3 = 35 \sin \theta$$

$$\dot{y} = -10t + 35 \sin \theta$$

integrate wrt x

$$y = -\frac{10t^2}{2} + 35t \sin \theta + C_4$$

$t=0$ $y=0$ $C_4=0$

$$y = -5t^2 + 35t \sin \theta$$

✓

ii) $x=50$ $y=15$

$$t = \frac{x}{35 \cos \theta}$$

$$= \frac{50}{35 \cos \theta}$$

$$= \frac{10}{7 \cos \theta}$$

$$y = 35t \sin \theta - 5t^2$$

$$15 = \frac{35^5 \times 10 \sin \theta}{7 \cos \theta} - \frac{5 \times 10^2}{7^2 \cos^2 \theta}$$

✓

$$15 = 50 \tan \theta - \frac{500}{49} (1 + \tan^2 \theta)$$

Q14 a) ii) continued.

$$15 \times 49 = 49 \times 50 \tan \theta - 500 - 500 \tan^2 \theta$$

$$500 \tan^2 \theta - 2450 \tan \theta + 1235 = 0 \quad \checkmark$$
$$\div 5$$

$$100 \tan^2 \theta - 490 \tan \theta + 247 = 0$$

$$\tan \theta = \frac{+490 \pm \sqrt{(490)^2 - 4 \times 100 \times 247}}{200} \quad \checkmark$$

$$\tan \theta \doteq 4.33$$

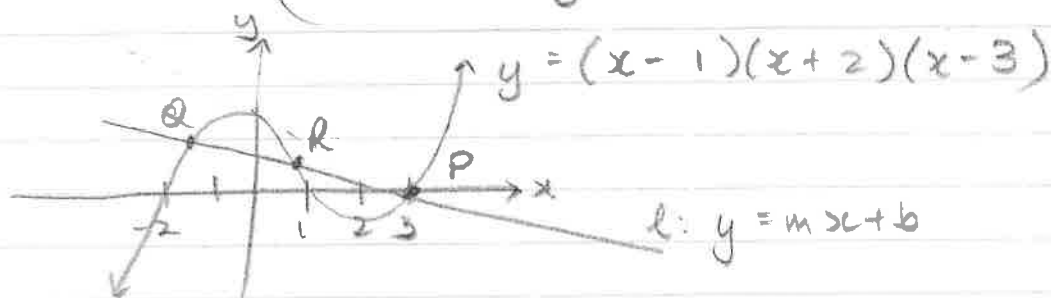
$$\tan \theta \doteq 0.571$$

$$\theta \doteq 77^\circ$$

$$\theta \doteq 30^\circ \quad \checkmark$$

(nearest degree)

b)



$P(3, 0)$ lies on l so $0 = 3m + b$
 $b = -3m$

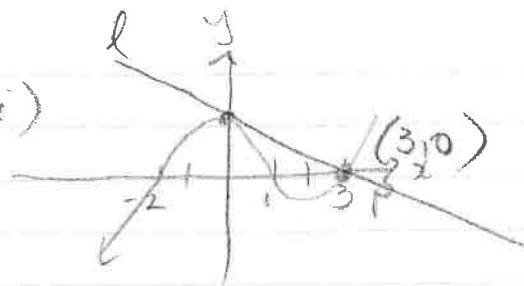
$l: y = mx - 3m$
intersection of curves

$$mx - 3m = x^3 - 2x^2 - 5x + 6$$

$$x^3 - 2x^2 - 5x - mx + 6 + 3m = 0 \quad \checkmark$$

$$x^3 - 2x^2 - (5+m)x + 6 + 3m = 0$$

Q14 b) ii)



let there be a double root α
represent the tangency.

so the roots are α , α and 3

$$\alpha + \alpha + 3 = 2$$

$$2\alpha = -1$$

$$\alpha = -\frac{1}{2} \quad \checkmark$$

$$3\alpha + 3\alpha + \alpha^2 = -(m+5)$$

$$6\alpha + \alpha^2 = -(m+5)$$

$$3 \times -\frac{1}{2} + \frac{1}{4} = -m-5$$

$$-2\frac{3}{4} = -m-5$$

$$m = -5 + 2\frac{3}{4}$$

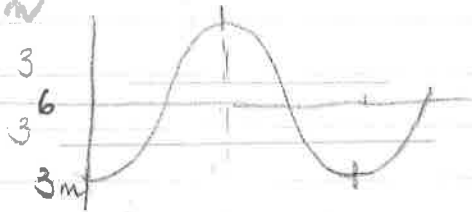
$$m = -\frac{9}{4} \quad \checkmark$$

$$y = -\frac{9}{4}x - 3 \times -\frac{9}{4}$$

$$4y = -9x + 27$$

$$\underline{9x + 4y - 27 = 0} \quad \checkmark$$

Q14 c) ^{1pm} high tide 9m



same low tide

San 1pm 9pm

centre of motion 6m

low tide to high tide = 8h

1 period is 16h

$$\rightarrow T = \frac{2\pi}{\frac{\pi}{8}} = \frac{16}{1} = 16$$

$$x = 6 - 3 \cos\left(\frac{\pi t}{8}\right) \quad \checkmark$$

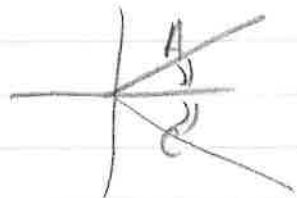
$$4 = 6 - 3 \cos \frac{\pi t}{8}$$

$$-2 = -3 \cos \frac{\pi t}{8}$$

$$\cos \frac{\pi t}{8} = \frac{2}{3}$$

$$\frac{\pi t}{8} = \cos^{-1}\left(\frac{2}{3}\right)$$

$$\frac{\pi t}{8} \doteq 0.841 \quad \checkmark$$



$$t \doteq \frac{0.841 \times 8}{\pi}$$

$$\doteq 2.14 \text{ h}$$

$$\frac{\pi t}{8} \doteq 2\pi - 0.841$$

$$t = \frac{(2\pi - 0.841) \times 8}{\pi}$$

$$\doteq 13.86 \text{ h}$$

The ship can pass safely
between 7:08am and 6:51pm ✓

* On that certain day, the ship could also pass safely from 12 midnight to 2:51am

$$Q14 d) \quad 2 \sin 3x \cos 4x - 1 - (\cos 4x + 2 \sin 3x) = 0$$

$$2 \sin 3x (\cos 4x + 1) - 1 (\cos 4x + 1) = 0 \quad \checkmark$$

$$(2 \sin 3x - 1) (\cos 4x + 1) = 0$$

$$\text{If } 2 \sin 3x = 1$$

$$\sin 3x = \frac{1}{2}$$

$$3x = n\pi + (-1)^n \sin^{-1}\left(\frac{1}{2}\right)$$

$$3x = n\pi + (-1)^n \frac{\pi}{6}$$

$$x = \frac{n\pi}{3} + (-1)^n \frac{\pi}{18}$$

$$\text{If } \cos 4x = -1$$

$$4x = 2n\pi \pm \cos^{-1}(-1)$$

$$4x = 2n\pi \pm \pi$$

$$x = \frac{n\pi}{2} \pm \frac{\pi}{4}$$

$$x = \frac{n\pi}{2} \pm \frac{\pi}{4}$$

$$x = \frac{n\pi}{3} + (-1)^n \frac{\pi}{18}$$

Looking for two smallest positive solutions.

$$\text{if } n = 0, 1$$

$$\text{if } n = 0, 1$$

$$x = 0 \pm \frac{\pi}{4}, \frac{\pi}{2} \pm \frac{\pi}{4}$$

$$x = 0 + \frac{\pi}{18}, \frac{\pi}{3} - \frac{\pi}{18}$$

$$x = \frac{\pi}{4 \times 18} + \frac{3\pi}{4}$$

$$x = \frac{\pi}{18}, \frac{5\pi}{18 \times 4} = \frac{20\pi}{72}$$

$$= \frac{18\pi}{72}$$

The two smallest positive solutions are $x = \frac{\pi}{18}$ and $x = \frac{\pi}{4}$ ✓